Gene flux erodes the karyotype barrier

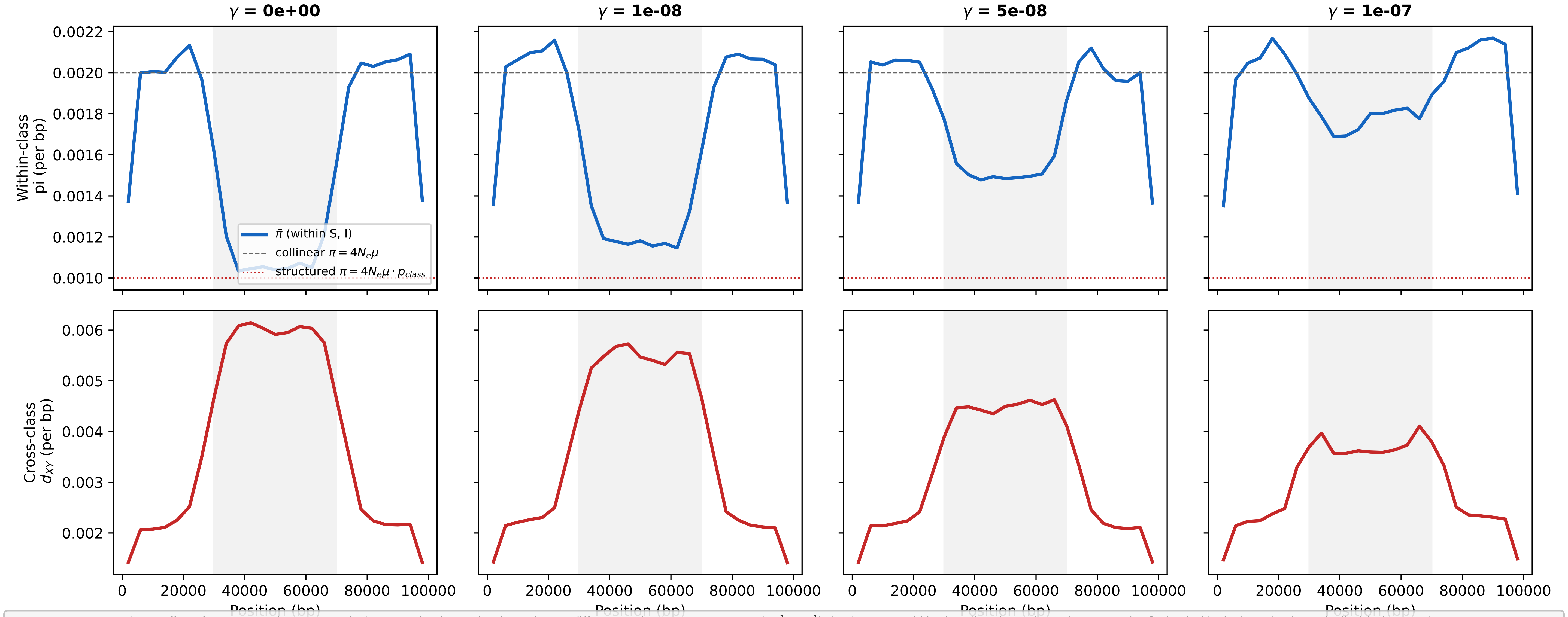

Figure. Effect of gene-conversion rate γ on the karyotype barrier. Each column shows a different γ value (0, 1e-8, 5e-8, 1e-7 bp^{-1} gen^{-1}). (Top) Average within-class diversity $\bar{\pi} = (\pi_S + \pi_I)/2$. At γ=0 (no flux), $\bar{\pi}$ inside the inversion (grey shading) is depressed to $4N_e\mu \cdot p_{class}$ (red dashed) — the structured-coalescent prediction. As γ increases, gene flux relaxes the class barrier and $\bar{\pi}$ recovers toward the collinear expectation $4N_e\mu$ (grey dashed), first at the centre (where ϕ(x) peaks) and last at breakpoints. (Bottom) Cross-class d_{XY} (S vs I). At γ=0, d_{XY} is uniformly elevated; as γ grows, the centre collapses first, leaving breakpoints as the last barrier. Parameters: Ne=50,000, p_inv=0.5, t_inv=200,000 gen, L=100 kb, μ=1e-8, r=1e-8, n_S=n_I=8, 40 replicates.
Command: pixi run -e all python examples/gene_flux_demo.py